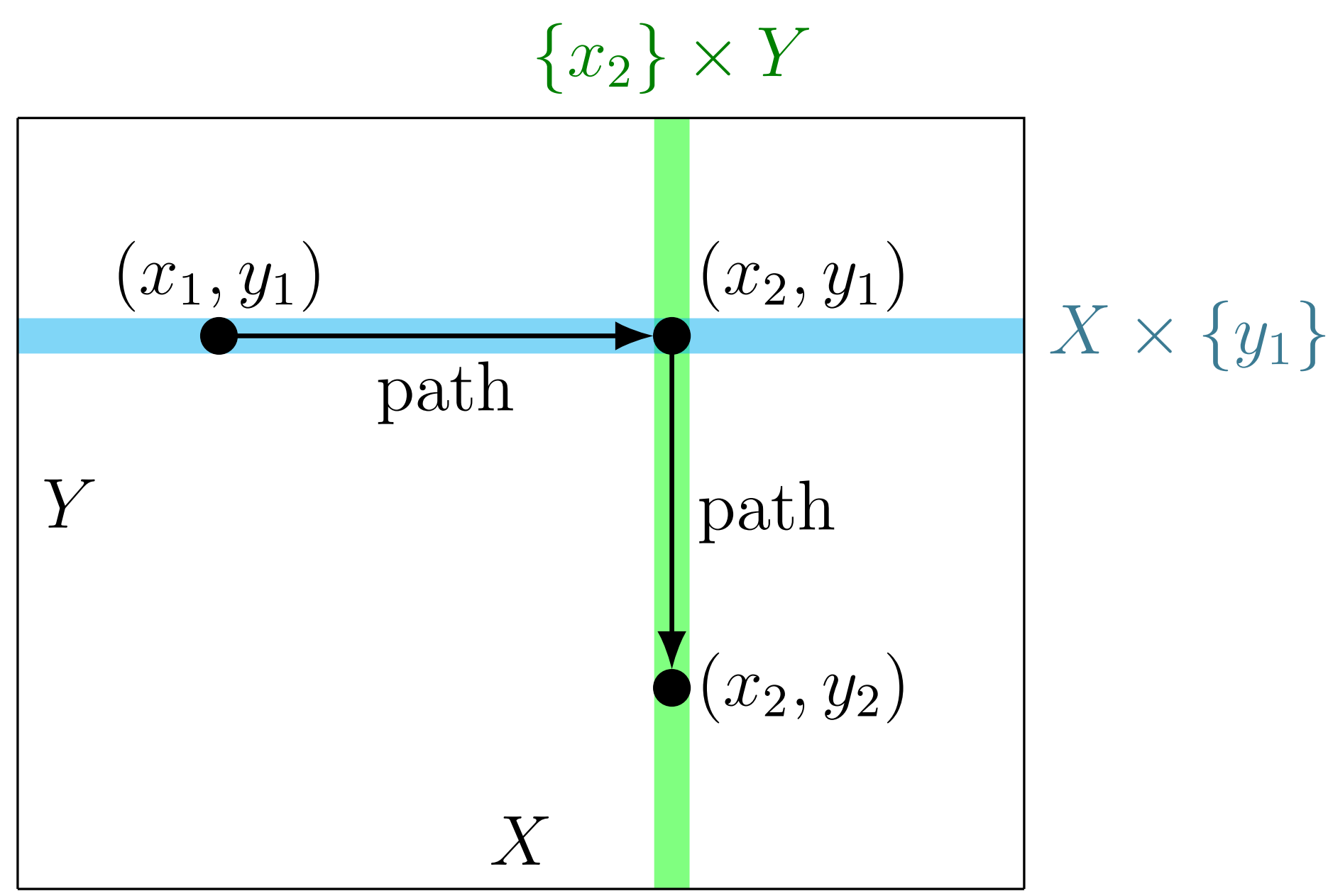
$\{x_2\} \times Y$
(x_1, y_1)
(x_2, y_1)
$X \times \{y_1\}$
path
Y
path
(x_2, y_2)
X